Assignment 1

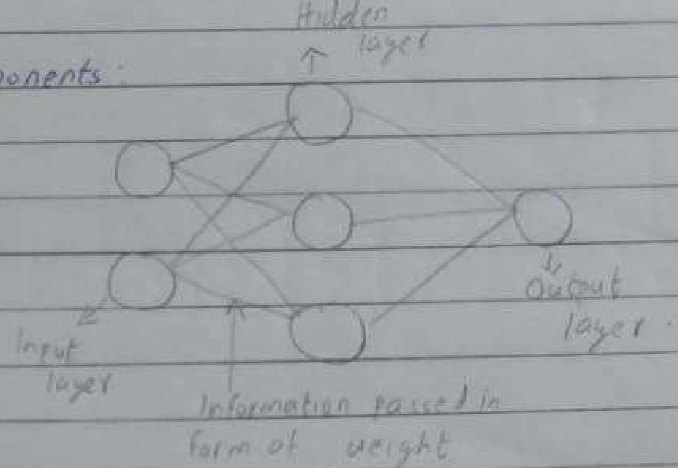
1. What is artificial Neural Network? Explain how it works?

→ The term 'artificial neural network' is derived from biological neural networks that develop the structure of a human brain. It consists of interconnected nodes (neurons) organized into layers.

Architecture:

It consist of 3 components:

- Input layer
- Hidden layer
- Output layer.



Working of ANN:

① In the first step, input unit are passed ie data is passed with some weights attached to it to the hidden layer. We can have any number of hidden layers.
Inputs $x_1, x_2, \ldots \ldots x_n$ is passed.

② Each hidden layer consists of neurons. All the inputs are connected to each neurons.

③ After passing on the inputs, all the computation is performed in the hidden layer. Computation performed in hidden layers are done in 2 steps which are as follows:

- First of all, all the input are multiplied by their weights. Weight is the gradient or coefficient of each variable. It shows the strength of the particular input. After assigning the weights a bias variable is added. Bias is a constant that helps the model to fit in the best way possible.

$$Z_1 = w_1 * x_1 + w_2 * x_2 + w_3 * x_3 + w_4 * x_4 + w_5 * x_5 + b$$

$w_1, w_2 \ldots w_5$ are the weights assigned to the input $x_1, x_2 \ldots x_5$ and b is the bias.

- Then is the second step, the activation function is applied to the linear equation Z. The activation function is a nonlinear transformation that is applied to the input before sending it to the next layer of neurons. The importance of the activation function is to inculcate nonlinearity in the model.

④ The whole process described in point ③ is performed in each hidden layer. After passing through every hidden layer we move to the last layer ie our output layer which gives us the final output. The process explained above is known as forwarding propagation

⑤ After getting the predictions from the output layer, the error is calculated ie the difference between the actual & the predicted outputs.

If the error is large then the steps are taken to minimize the errors & for the same purpose Back Propagation is performed.

2. Explain working of Biological and Artificial Neurons with proper labelled images.

→

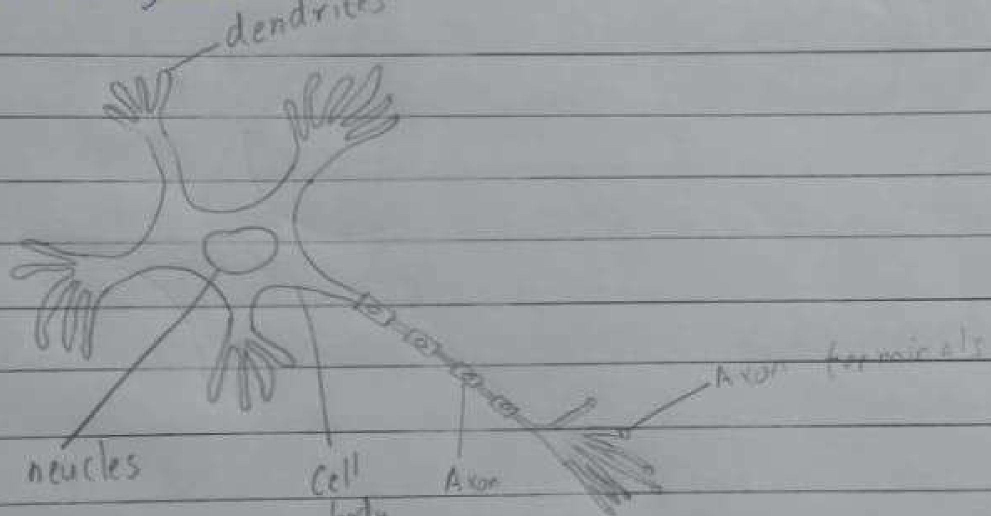
Biological Neurons:

Neurons are the basis functional units of the nervous system and they generate electrical signals called action potentials, which allows them to quickly transmit information over long distances. Almost all the neurons has 3 basic functions essential for normal functioning of all the cells in the body. These are to

① Perceive Singnal (or information) from outside)

② Process the incoming signals & determine whether or not the information should be passed along.

③ Communicate signals to target cells which might be other neurons or muscles or glands.



Basic parts of neurons:

① Dendrite

- Dentrite are responsible for getting incoming signals from outside
- Accepts input (signal) from other neurons (electric impulses).

② Soma

- Soma is the cell body responsible for the processing of input signals and deciding whether a neuron should fire an output signal.
- It sums all the incoming signal to generate input

③ Axon

- Axon is responsible for getting processed signals from neuron to relevant cells.
- When the sum reaches a threshold value, neuron fires and the signal travels down the axon to the other neurons. ~~a~~ Convert processed inputs into output.

④ Synapse

- Synapse is the connection between an axon and other neuron dendrites.
- The electrochemical contact between neurons. Using synapse a neuron can transfer the output of that neuron to the input of the next neuron.

Artificial Neurons:

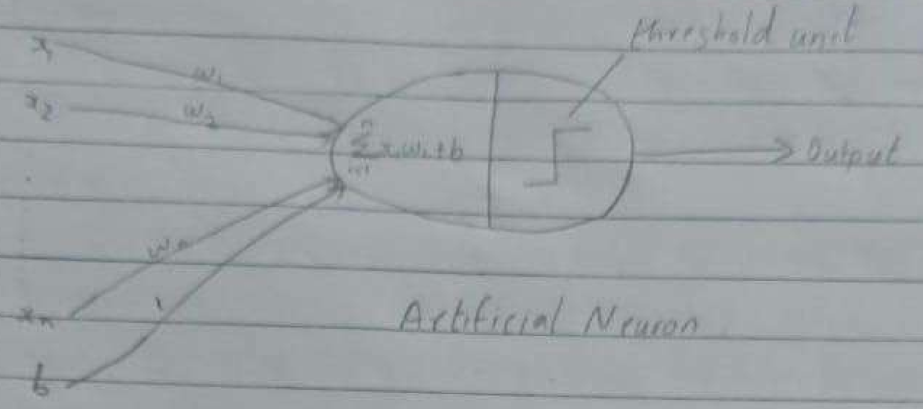
Artificial neuron also known as perception is the basic unit of the neural network. In simple terms, it is a mathematical function based on a model of biological neurons.

Each artifical neuron has the following main function

① Takes inputs from the input layer

② Weighs them separately & sums them up

③ Pass this sum through a nonlinear function to produce output.



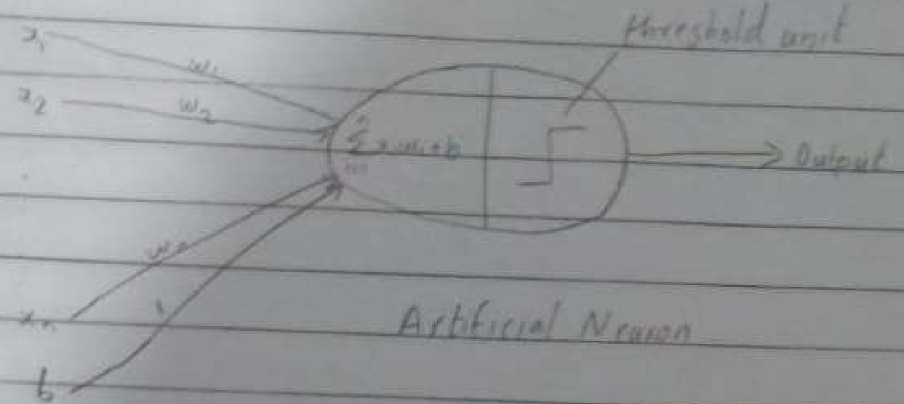
Artificial Neuron

$w_1, w_2 \cdots w_n \rightarrow$ weight of connection, $x_1 x_2 \cdots x_n \rightarrow$ Input, $b \rightarrow$ bias

The perceptron (neuron) consist of 4 parts:

1. Input values or one input layer
2. Weight & bias
3. Activation function
4. Output layer

Biological Neuron	Artificial Neuron
Dendrites	Input
Cell Nucleus	Node
Axon	Output
Synapse	Interconnections.



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3. What is Activation function and why do we need it?

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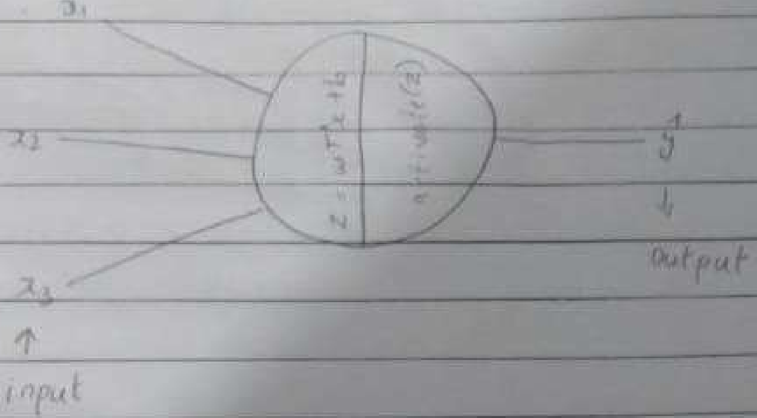
The activation function is a mathematical "gate" in between the input feeding the current neuron and its output going to the next layer. They basically decide whether the neuron should be activated or not.

eg:



Without activation function, weight and bias would ~~ob~~ only have a linear transformation or neural network is just a linear regression model, a linear equation is polynomial of one degree only which is simple to solve but limited in terms of ability to solve complex problems or higher degree polynomials But opposite to that, the purpose of an activation function is to add non-linearity to the neural network.

They basically decide to deactivate neurons or activate them to get the desired output. It also performs a non linear transformation on the input to get better results on a complex neural network.

Activation function also helps to normalize the output of any input in the range between 1 to -1. Activation function must be efficient and it should reduce the computation time because the neural network sometimes trained on millions of data points.

4. Explain different types of activation function.

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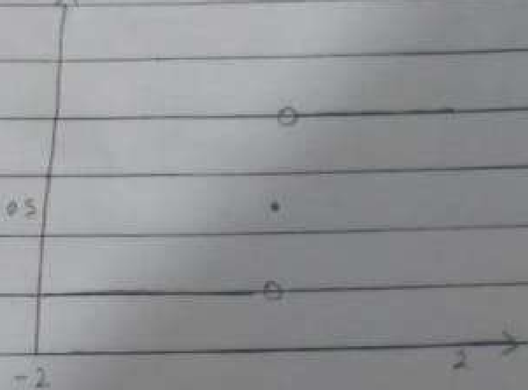
- Binary Step function.

It depends on a threshold value that decides whether a neuron should be activated or not.



The input fed to the activation function is compared to a certain threshold; if the input is greater than it then the neuron is activated else it is deactivated, meaning that it's output is not passed on to the next hidden layer.

Mathematically it can be represented as,

$$f(x) = \begin{cases} 0 & , \text{ for } x < 0 \\ 1 & , \text{ for } x \geq 0 \end{cases}$$

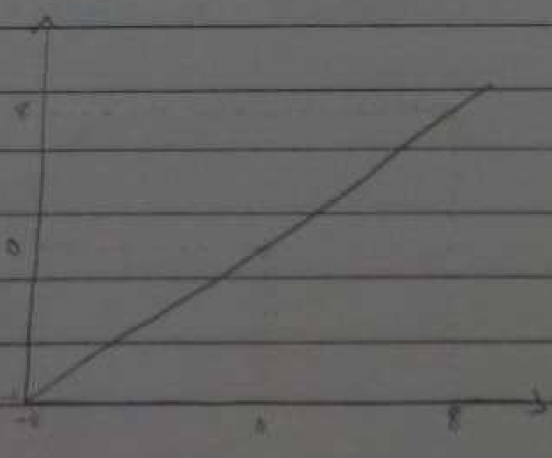
Limitations:

- It cannot provide multi-value output. For example, it cannot be used for multi-class classification problem.
- The gradient of the step function is zero, which causes a hindrance in the backpropagation process.

- Linear Activation function

It is also known as identity function where the activation is proportional to the output.

Mathematically it can be represented as, $f(x) = x$



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limitations.
- It's not possible to use backpropagation as the derivative of the function is a constant and has no relation to the input x.
- All layer of the neural network will collapse into one if a linear activation function is used. No matter the number of layers in the neural network, the last layer will still be linear function of the first layer. So essentially, a linear activation function turn the neural network into just one layer.

o Non-linear Activation function.
Non-linear activation function solve the following limitation of linear activation function:
- They allow back propagation because now the derivative function would be related to the input and it's possible to go back and understand which weight in the input neurons can provide a better prediction.
- They allow the stacking of multiple layers of neurons as the output would now be a non-linear combination of input passed through multiple layers. Any output can be represented as a functional computation in a neural network.

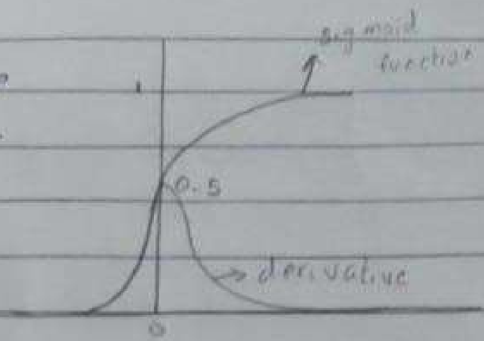
① Sigmoid / Logistic Activation function
The sigmoid function is an activation function where it scales the values between 0 and 1 by applying a threshold.

Mathematically it can be represented as

$$f(x) = \frac{1}{1+e^{-x}}$$

Uses:

- It is commonly used for models where we have to predict the probability as an output. Since probability of anything exist only between the range of 0 and 1, sigmoid is the right choice because of its range.



- The function is differentiable and provides a smooth gradient ie preventing jumps in output values. This is represented by an S-shaped of the sigmoid activation function.

limitation:

- The sigmoid output is close to zero is highly negative input. This can be a problem in neural network training and can lead a slow learning & the model getting trapped in local minima during training.
- As the gradient value approaches zero, the network ceases to learn and suffer from vanishing gradient problem.
- Hence hyperbolic tangent is more preferable as an activation function in hidden layers of a neural network.

② Tanh function (Hyperbolic Tangent)

Tanh function is very similar to the sigmoid / logistic activation function, and even have the same s-shape with difference in output range of -1 to 1.

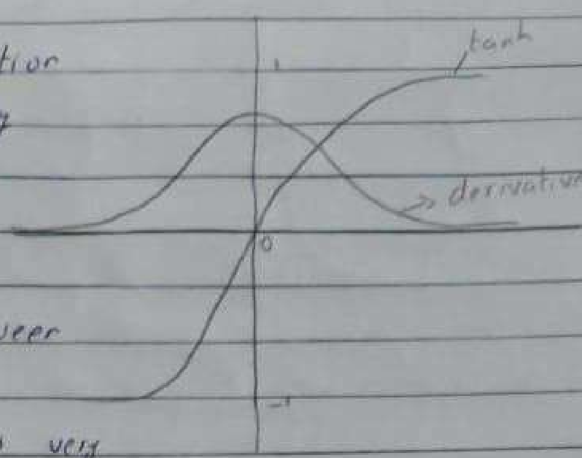
In Tanh the larger the input (more positive), the closer the output value will be to 1 whereas smaller the input (more negative) the closer the output will be to -1.

Mathematically it can be represented as

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Advantages:

- The output of the tanh activation function is zero centered, hence we can easily map the output values as strongly negative, neutral or strongly positive.
- Usually used in hidden layer of a neural network as its values lie between -1 to 1, therefore the mean for the hidden layer comes out to be 0 or very close to it. It helps in centering the data and makes learning for the next layer much easier



Limitations:

It also faces the problem of vanishing gradients similar to the sigmoid activation function. Plus the gradient of the tanh function is much steeper as compared to the sigmoid function.

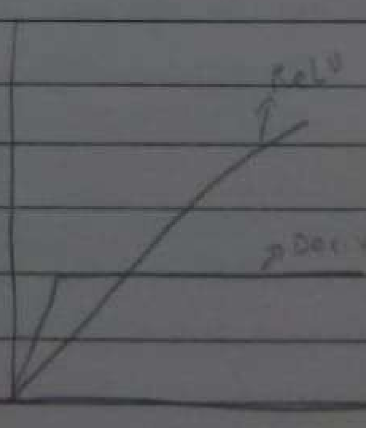
③ ReLU (Rectified Linear Unit)

This is one of the most widely used activation function. The benefits of ReLU is the sparasity, it allows only values which are positive & negative values are not passed which will speed up process. Mathematically it can be represented as.

$$f(x) = max(0, x)$$

Advantages:

- Since only a certain number of neurons are activated, the ReLU function is far more computationally efficient when compared to the sigmoid and tanh function.



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- ReLU accelerates the convergence of gradient descent towards the global minimum.

Limitation:

- The Dying ReLU problem
- The negative side of the graph makes the gradient value zero. Due to this reason, during the back propagation process, the weights and biases for some neurons are not updated. This can create dead neuron which never gets activated.
- All the negative input value become zero immediately which decreases the model's ability to fit or train from the data properly

(5) Softmax

It is often described as a combination of multiple signals. The softmax function can be used for multiclass classification problems. This function returns the probability for a datapoint belonging to each individual class.

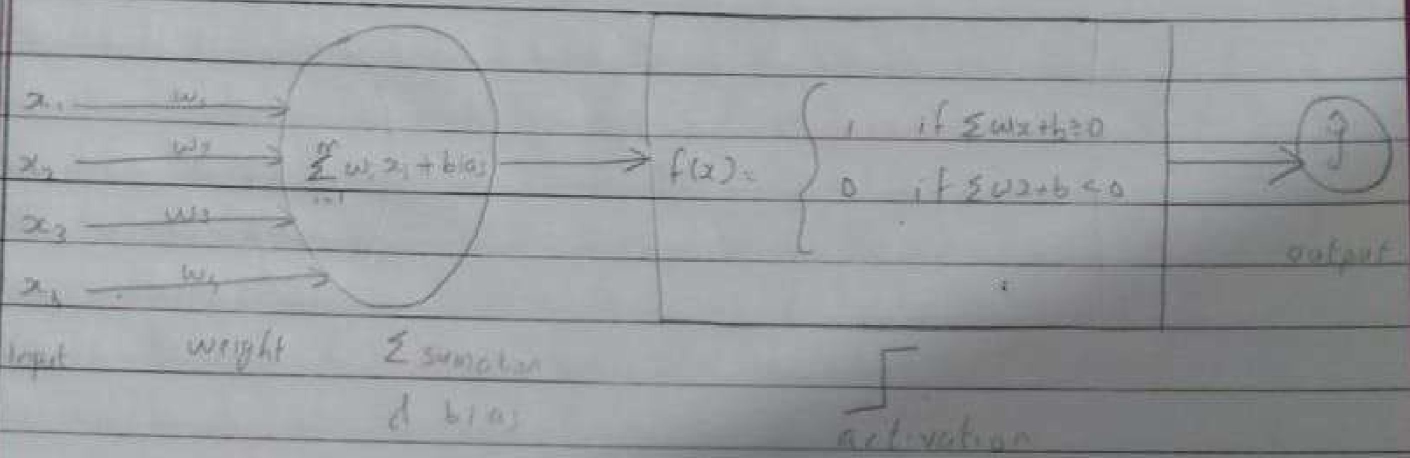
Mathematically it can be represented as

$$\text{Softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$

While building a network for a multiclass problem, the output layer would have as many neurons as the number of classes in the target.

Q5. Explain Single and Multi-layered perceptron with proper figure.
→

Single-layered perceptron models
A single-layer perceptron model includes a feed-forward network depends on a threshold transfer function in its model.
It is the easiest type of artificial neural network that able to analyze only linearly separable objects with binary outcomes (target) i.e. 1 & 0.



m = number of input to the perceptron
It can also be represented as 1 or -1 depending on which activation function is used.
Single-layered perceptron model's algorithm doesn't have previous information, so initially weight are allocated inconstantly then the algorithm adds up all the weight inputs, if the added value is more than some pre-determined value (or threshold value) then single layered perceptron is stated as activation & delivered output

$$f(x) = \begin{cases} 0 \\ 1 \end{cases}, \quad f'(x) = 0$$

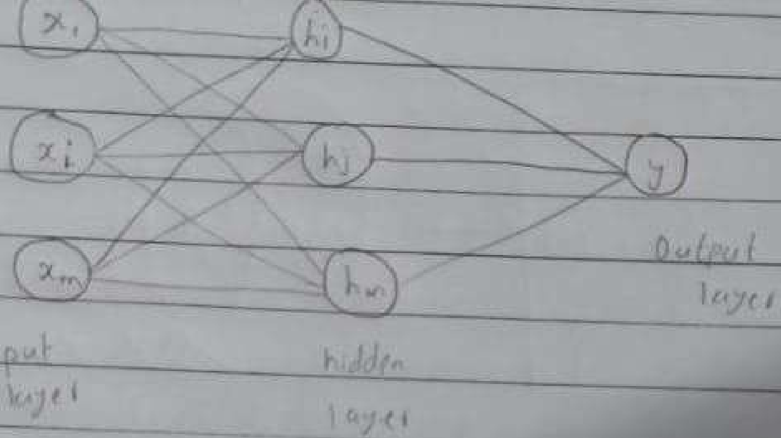
In single layer perceptron backward propagation is not possible.

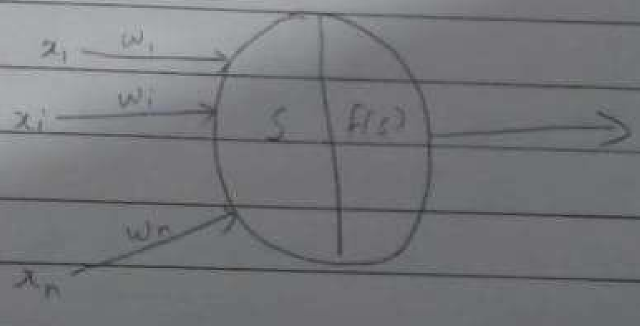
Multi - layered perceptron model

A multi - layered perceptron model has a structure similar to a single - layered perceptron model with more number of hidden layers. It is also termed as backpropagation algorithm.

It executes in 2 stages:

- The forward stage
- The backword stage.





S = Summation , f(s) = transformation.

Multilayer perceptrons or feed forward neural network with 2 or more layers have the greater processing power.

6. Sums.

i) Sigmoid activation derivative

$$\sigma = \frac{1}{1+e^{-x}}$$

$$\sigma' = \frac{0-(-e^{-x})}{(1+e^{-x})^2}$$

$$= \frac{e^{-x}}{1+e^{-x}} \cdot \frac{1}{1+e^{-x}}$$

$$= \frac{1}{1+e^{-x}} \left(\frac{1+e^{-x}-1}{1+e^{-x}} \right)$$

$$= \frac{1}{1+e^{-x}} \left(1 - \frac{1}{1+e^{-x}} \right)$$

$$= \sigma(1-\sigma)$$

ii) tanh Activation derivative

$$\sigma = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

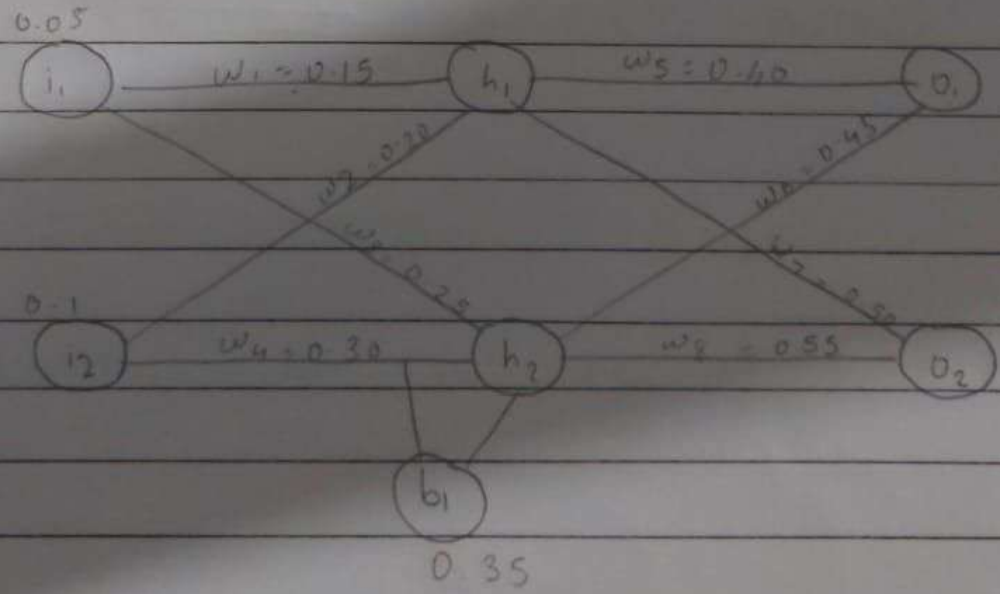
$$\sigma' = \frac{(e^{x} + e^{-x})(e^{x} + e^{-x}) - (e^{x} - e^{-x})(e^{x} - e^{-x})}{(e^{x} + e^{-x})^{2}}$$

$$= \frac{(e^{x} + e^{-x})^{2} - (e^{x} - e^{-x})^{2}}{(e^{x} + e^{-x})^{2}}$$

$$= 1 - \frac{(e^{x} - e^{-x})^{2}}{(e^{x} + e^{-x})^{2}}$$

$$= 1 - \left(\frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}\right)^{2}$$

$$= 1 - \sigma^{2}$$



Target values $T_1 = 0.01$, $T_2 = 0.99$; $\alpha = 0.$